

Tema 1 - Repaso

Ejercicios

Verificación si una función es solución

1. $y' = 3x^2$ $y = x^3 + 7$

$$(x^3 + 7)' = 3x^2 \rightarrow 3x^2 = 3x^2 \rightarrow y \text{ es solución}$$

2. $y' + 2y = 0$, $y = 3e^{-2x}$

$$(3e^{-2x})' + 2(3e^{-2x}) = 0 \rightarrow -6e^{-2x} + 6e^{-2x} = 0 \rightarrow y \text{ es solución}$$

3. $y'' + 4y = 0$, $y_1 = \cos 2x$ $y_2 = \sin 2x$

$$(\cos 2x)'' + 4\cos 2x = 0 \rightarrow -4\cos 2x + 4\cos 2x = 0 \rightarrow y_1 \text{ es solución}$$

$$(\sin 2x)'' + 4\sin 2x = 0 \rightarrow -4\sin 2x + 4\sin 2x = 0 \rightarrow y_2 \text{ es solución}$$

4. $y'' = 9y$ $y_1 = e^{3x}$ $y_2 = e^{-3x}$

$$(e^{3x})'' = 9e^{3x} \rightarrow 9e^{3x} = 9e^{3x} \rightarrow y_1 \text{ es solución}$$

$$(e^{-3x})'' = 9e^{-3x} \rightarrow 9e^{-3x} = 9e^{-3x} \rightarrow y_2 \text{ es solución}$$

10. $x^2 y'' + xy' - y = \ln(x)$ $y_1 = x - \ln(x)$ $y_2 = \frac{1}{x} - \ln(x)$

$$y_1' = 1 - \frac{1}{x}$$

$$y_2' = \frac{1}{x^2} - \frac{1}{x}$$

$$y_1'' = \frac{1}{x^2}$$

$$y_2'' = \frac{1}{x^3} + \frac{1}{x}$$

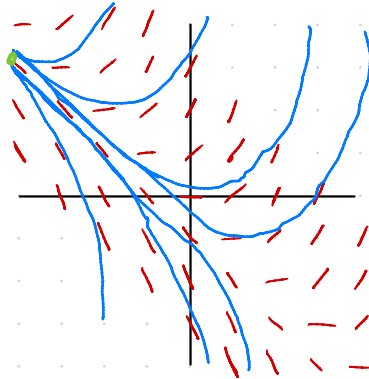
$$\frac{1}{x^2} \cdot x^2 + x \left(1 - \frac{1}{x}\right) - x + \ln(x) = \ln(x) \rightarrow \cancel{x^2} + \cancel{x^2} - \cancel{x} - \cancel{x} + \ln(x) = \ln(x) \rightarrow y_1 \text{ es solución}$$

$$\left(\frac{1}{x^3} + \frac{1}{x^2}\right)x^2 + x\left(-\frac{1}{x^2} - \frac{1}{x}\right) - \frac{1}{x} + \ln(x) = \ln(x) \rightarrow \frac{1}{x} + 1 - \frac{1}{x} - 1 - \frac{1}{x} + \ln(x) \neq \ln(x) \text{ y no es solución}$$

Campos de Isoclinas y Curvas Solución

21. $y' = x + y \rightarrow y = y' - x$

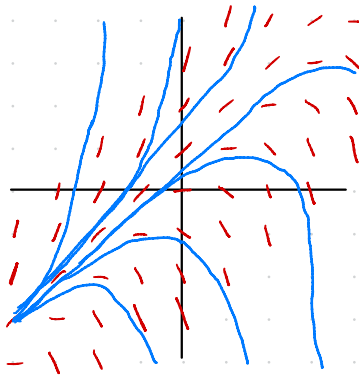
c	$f(x)$
0	$y = -x$
1	$y = 1 - x$
-1	$y = -1 - x$
2	$y = 2 - x$
-2	$y = -2 - x$



$y(-4) \approx 3$

22. $y' = y - x \rightarrow y = y' + x$

c	$f(x)$
0	$y = x$
1	$y = 1 + x$
-1	$y = -1 + x$
2	$y = 2 + x$
-2	$y = -2 + x$



$y(-4) \approx -3$

Existencia y unicidad de las soluciones

11. $\frac{dy}{dx} = 2x^2y^2 \rightarrow \frac{dy}{dx}$ es siempre continua $\rightarrow$ Hay existencia
 $f(x,y) = 2x^2y^2 \quad \frac{\partial f}{\partial y} = 2x^2y \rightarrow \frac{\partial f}{\partial y}$ es continua $\rightarrow$ Hay unicidad en $x=1$
12. $\frac{dy}{dx} = x \ln y \rightarrow \frac{dy}{dx}$ continua $\forall y > 0 \rightarrow y=1 \rightarrow$ Hay Existencia en $x=1$
 $\frac{\partial f}{\partial y} = \frac{x}{y} \rightarrow$ unicidad $\forall y \neq 0 \rightarrow y=1 \rightarrow$ Hay unicidad en $x=1$
13. $\frac{dy}{dx} = \sqrt[3]{y} \quad y(0)=1 \rightarrow \frac{dy}{dx}$ es continua siempre $\rightarrow$ Hay existencia en $x=0$
 $\frac{\partial f}{\partial y} = \frac{1}{3} \frac{1}{(\sqrt[3]{y})^2} \rightarrow$ unicidad si $y \neq 0 \rightarrow$ Hay unicidad en $x=0$
14. $\frac{dy}{dx} = \sqrt[3]{y} \quad y(0)=0 \rightarrow$ como mencionado en 13) $\rightarrow$ Hay existencia en $x=0$
 $\frac{\partial f}{\partial y} = \frac{1}{3} \frac{1}{(\sqrt[3]{y})^2} \rightarrow$ unicidad si $y \neq 0 \rightarrow$ No hay unicidad ya que $y=0$
15. $\frac{dy}{dx} = \sqrt{x-y} \quad y(2)=2 \rightarrow$ Existencia si $x \geq y \rightarrow$ Hay existencia en $x=2$
 $\frac{\partial f}{\partial y} = -\frac{1}{2} \frac{1}{\sqrt{x-y}} \rightarrow$ unicidad si $x > y \rightarrow$ No hay unicidad en $x=2$
17. $\frac{dy}{dx} = \frac{x-1}{y} \rightarrow$ Existencia en $x=0$ ($y \neq 0$)
 $\frac{\partial f}{\partial y} = -\frac{x-1}{y^2} \rightarrow$ unicidad en $x=0$ ($y \neq 0$)
16. $\frac{dy}{dx} = \frac{x-1}{y} \rightarrow$ No hay existencia ni unicidad en $x=1$ ($y=0$)

Ecuaciones Separables y Aplicaciones

$$1. \frac{dy}{dx} + 2xy = 0 \rightarrow \frac{dy}{dx} = -2xy \rightarrow \frac{1}{y} \frac{dy}{dx} = -2x \rightarrow \int \frac{1}{y} dy = \int -2x dx$$

$$\ln y = -x^2 + C \rightarrow y = e^{-x^2+C} = Ae^{-x^2}$$

$$2. \frac{dy}{dx} = -2xy^2 \rightarrow \frac{1}{y^2} \frac{dy}{dx} = -2x \rightarrow \int \frac{1}{y^2} dy = \int -2x dx \rightarrow \frac{1}{y} = x^2 + C \rightarrow y = \frac{1}{x^2 + C}$$

$$10. (1+x)^2 \frac{dy}{dx} = (1+y)^2 \rightarrow \frac{1}{(1+y)^2} \frac{dy}{dx} = \frac{1}{(1+x)^2} \rightarrow \int \frac{1}{(1+y)^2} dy = \int \frac{1}{(1+x)^2} dx$$

$$-\frac{1}{1+y} = -\frac{1}{1+x} \rightarrow \frac{1}{1+y} = \frac{1}{1+x} - \frac{C+C}{x+1} \rightarrow 1+y = \frac{1+x}{-C-x-C+1} \rightarrow y = \frac{1+x}{-C-x-C+1} + 1$$

$$12. yy' = x(y^2+1) \rightarrow \int \frac{y}{1+y^2} dy = \int x dx \rightarrow \frac{1}{2} \ln(1+y^2) = \frac{x^2}{2} + C \rightarrow \ln(1+y^2) = x^2 + 2C \rightarrow 1+y^2 = Ae^{x^2}$$

$$y^2 = Ae^{x^2} - 1$$

$$15. \frac{dy}{dx} = \frac{(x-1)y^5}{x^2(1+y^2)} \rightarrow \frac{y(1+y^2-1)}{y^5} dy = \frac{(x-1)}{x^2} dx$$

$$\int \frac{(1+y^2-1)}{y^4} dy = \int \frac{(x-1)}{x^2} dx$$

$$\int \frac{(x-1)}{x^2} = \int \frac{x}{x^2} - \frac{1}{x^2} dx = \int \frac{1}{x} - \frac{1}{x^2} dx = \ln(x) + \frac{1}{x} = \frac{x \ln(x) + 1}{x} + C$$

$$\int \frac{(1+y^2-1)}{y^4} dy = \int \frac{2}{y^4} - \frac{1}{y^4} dy = -\frac{2}{y^3} + \frac{1}{3} \frac{1}{y^3}$$

$$-\frac{2}{y^3} + \frac{1}{3y^3} = \ln(x) + \frac{1}{x} + C$$

$$17. y' = 1+x+y+xy \rightarrow (1+x) + (1+x)y = \frac{dy}{dx} \rightarrow \frac{dy}{dx} = (1+x)(1+y) \rightarrow \int \frac{1}{1+y} dy = \int (1+x) dx$$

$$\ln(1+y) = x + \frac{x^2}{2} + C \rightarrow 1+y = Ae^{x^2/2+x} \rightarrow y = Ae^{x^2/2+x} - 1$$

Factor Integrante

$$3. y' + 3y = 2xe^{-3x} \quad \begin{cases} P(x) = 3 \\ Q(x) = 2xe^{-3x} \end{cases}$$

$$\rho(x) = e^{\int 3 dx} = e^{3x}$$

$$e^{3x} y' + 3e^{3x} y = 2xe^{-3x} e^{3x} \rightarrow \frac{d(e^{3x} y)}{dx} = 2x \rightarrow e^{3x} y = \int 2x = x^2 + C$$
$$y = e^{-3x} (x^2 + C)$$

$$21. xy' = 3y + x^4 \cos(x) \quad y(2\pi) = 0$$

$$y' - \frac{3}{x} y = x^3 \cos(x) \quad \begin{cases} P(x) = -3/x \\ Q(x) = x^3 \cos(x) \end{cases}$$

$$\rho(x) = e^{\int -3/x dx} = e^{-3 \ln(x)} = x^{-3} = \frac{1}{x^3}$$

$$\frac{1}{x^3} y' - \frac{3}{x^4} y = \cos(x) \rightarrow \frac{1}{x^3} y = \int \cos(x) dx \rightarrow \frac{y}{x^3} = \sin(x) + C \rightarrow y = x^3 \sin(x) + Cx^3$$

$$y(2\pi) = 0 \rightarrow 0 = \sin(2\pi) + C \rightarrow C = 0 \rightarrow y = x^3 \sin(x)$$

$$8. 3xy' + y = 12x \rightarrow y' + \frac{1}{3x} y = 4 \quad \begin{cases} P(x) = \frac{1}{3x} \\ Q(x) = 4 \end{cases}$$

$$\rho(x) = e^{\int \frac{1}{3x} dx} = e^{\frac{1}{3} \ln(x)} = x^{1/3}$$

$$x^{1/3} y' + \frac{1}{3} x^{-2/3} y = 4x^{1/3} \rightarrow \frac{d(x^{1/3} y)}{dx} = 4x^{1/3} \rightarrow x^{1/3} y = \int 4x^{1/3} dx = 3x^{4/3} + C$$

$$y = \frac{C}{x^{1/3}} + \frac{3x^{4/3}}{x^{1/3}} = \frac{C}{\sqrt[3]{x}} + 3x$$

$$22. y' = 2xy + 3x^2 e^{x^2} \quad y(0) = 5$$

$$y' - 2xy = 3x^2 e^{x^2} \quad \begin{cases} P(x) = -2x \\ Q(x) = 3x^2 e^{x^2} \end{cases}$$

$$\rho(x) = e^{\int -2x dx} = e^{-x^2}$$

$$e^{-x^2} y' - 2x e^{-x^2} y = 3x^2 \rightarrow e^{-x^2} y = \int 3x^2 dx = x^3 + C \rightarrow y = e^{x^2} (x^3 + C)$$

$$y(0) = 5 \rightarrow 5 = C \rightarrow y(x) = e^{x^2} (x^3 + 5)$$

$$16. y' = (1-y) \cos(x) \rightarrow y(\pi) = 2$$

$$y' = \cos(x) - y \cos(x) \rightarrow y' + \cos(x)y = \cos(x) \quad \begin{cases} P(x) = \cos(x) \\ Q(x) = \cos(x) \end{cases}$$

$$P(x) = e^{\int \cos(x)} = e^{\sin(x)} \rightarrow e^{\sin(x)} y' + \cos(x) e^{\sin(x)} y = \cos(x) e^{\sin(x)}$$

$$e^{\sin(x)} y = \int \cos(x) e^{\sin(x)} dx = e^{\sin(x)} + C \rightarrow y(x) = 1 + \frac{C}{e^{\sin(x)}} \quad y(\pi) = 2$$

$$2 = 1 + \frac{C}{1} \rightarrow C = 1 \rightarrow y(x) = 1 + \frac{1}{e^{\sin(x)}}$$

$$13. y' + y = e^x \quad \begin{cases} P(x) = 1 \\ Q(x) = e^x \end{cases} \quad y(0) = 1$$

$$P(x) = e^{\int 1 dx} = e^x$$

$$e^x y' + e^x y = e^{2x} \rightarrow e^x y = \int e^{2x} dx = \frac{1}{2} e^{2x} + C \rightarrow y = \frac{1}{2} \frac{e^{2x}}{e^x} + \frac{C}{e^x} = \frac{1}{2} e^x + \frac{C}{e^x}$$

$$y(0) = 1 \quad 1 = \frac{1}{2} e^0 + C \rightarrow C = \frac{1}{2} \rightarrow y(x) = \frac{1}{2} e^x + \frac{1}{2e^x} = \frac{e^x + e^{-x}}{2} = \cosh(x)$$

$$25. (x^2+1)y' + 3x^3y = 6x e^{-\frac{3}{2}x^2} \quad y(0) = 1$$

$$y' + \frac{3x^3}{x^2+1} y = \frac{6x}{x^2+1} e^{-\frac{3}{2}x^2} \quad \begin{cases} P(x) = \frac{3x^3}{x^2+1} \\ Q(x) = \frac{6x}{x^2+1} e^{-\frac{3}{2}x^2} \end{cases}$$

$$x^2+1 \left[\frac{3x^3}{3x^3 + 3x} \right] \rightarrow \frac{3x^3}{x^2+1} = 3x - \frac{3x}{x^2+1}$$

$$P(x) = e^{\int 3x - \frac{3x}{x^2+1}} = e^{\frac{3}{2}x^2 - \frac{3}{2} \ln(x^2+1)} = e^{\frac{3}{2}x^2} \cdot (x^2+1)^{-\frac{3}{2}}$$

$$e^{\frac{3}{2}x^2} (x^2+1)^{-\frac{3}{2}} y' + 3x^3 (e^{\frac{3}{2}x^2} (x^2+1)^{-\frac{3}{2}}) y = \frac{6x}{(x^2+1)} \cdot (x^2+1)^{-\frac{3}{2}} = \frac{6x}{(1+x^2)^{\frac{5}{2}}}$$

$$e^{\frac{3}{2}x^2} (x^2+1)^{-\frac{3}{2}} y = \int \frac{6x}{(x^2+1)^{\frac{5}{2}}} dx = \int 6x (x^2+1)^{-\frac{5}{2}} dx = -2(x^2+1)^{-\frac{3}{2}} + C$$

$$y(x) = \frac{2e^{-\frac{3}{2}x^2}}{(x^2+1)^{\frac{3}{2}}} + \frac{C(x^2+1)^{\frac{3}{2}}}{e^{\frac{3}{2}x^2}} \rightarrow 1 = C - 2 \rightarrow C = 3 \quad y(x) = -\frac{2}{e^{\frac{3}{2}x^2}} + \frac{3(x^2+1)^{\frac{3}{2}}}{e^{\frac{3}{2}x^2}}$$

Sustitución y Diferenciales Exactas

14. $x e^y y' = 2e^y + 2x^2 e^{2x} \rightarrow y' = \frac{2}{x} + \frac{2x^2 e^{2x}}{e^y} \quad v = e^y \rightarrow y = \ln(v)$

$$\frac{dy}{dx} = P_x + P_v \frac{dv}{dx} = \frac{1}{v} \frac{dv}{dx}$$

$$\frac{1}{v} \frac{dv}{dx} = \frac{2}{x} + \frac{2x^2 e^{2x}}{v} \rightarrow \frac{dv}{dx} = \frac{2}{x} v + 2x^2 e^{2x} \rightarrow \frac{dv}{dx} - \frac{2}{x} v = 2x^2 e^{2x} \quad \begin{cases} P(x) = -\frac{2}{x} \\ Q(x) = 2x^2 e^{2x} \end{cases}$$

$$P(x) = e^{\int -\frac{2}{x} dx} = e^{-2 \ln(x)} = x^{-2}$$

$$\frac{1}{x^2} \frac{dv}{dx} - \frac{2}{x} \frac{1}{x^2} v = \frac{2x^2 e^{2x}}{x^2} \rightarrow \frac{1}{x^2} v' - \frac{2}{x^3} v = 2e^{2x} \rightarrow \frac{1}{x^2} v = \int 2e^{2x} = e^{2x} + C$$

$$v(x) = x^2 e^{2x} + Cx^2 \rightarrow v = e^y \rightarrow e^y = x^2 e^{2x} + Cx^2 \rightarrow y = \ln(x^2 e^{2x} + Cx^2)$$

15. $x(x+y)y' + y(3x+y) = 0$

$$x(x+y)y' + 3xy + y^2 = 0 \rightarrow y' + \frac{3x}{x(x+y)} y + \frac{y^2}{x(x+y)} = 0 \rightarrow y' = -\frac{y(3x+y)}{x(x+y)} \quad y = vx \rightarrow v' = \frac{y'}{x}$$

$$y' = P_x + P_v \frac{dv}{dx} = v + x \frac{dv}{dx} \rightarrow v + x v' = -\frac{vx(3x+vx)}{x(x+vx)} = -\frac{v(3x+vx)}{x+vx} = -\frac{3vx+vx^2}{x+vx}$$

$$x v' = -\frac{3vx+vx^2}{x+vx} - v = -\frac{3vx+vx^2}{x+vx} - \frac{v(x+vx)}{x+vx} = \frac{-4vx-2v^2x}{x+vx} \rightarrow v' = \frac{-4v-2v^2}{x(1+v)} \rightarrow \frac{(1+v)}{-2(1+v^2)} v' = \frac{1}{x}$$

$$\int \frac{(1+v)}{-2(1+v^2)} dv = \int \frac{1}{x} dx \rightarrow -\frac{1}{2} \int \frac{1}{2} \frac{2(1+v)}{(1+v^2)} dv = \int \frac{1}{x} dx \rightarrow -\frac{1}{4} \ln(1+v^2) = \ln(x) + C$$

$$\ln(1+v^2) = -4 \ln(x) + C \rightarrow 1+v^2 = e^{\ln(x^{-4}) + C} = Cx^{-4} \rightarrow x^4(1+v^2) = C$$

$$x^4 \left(2\frac{y}{x} + \frac{y^2}{x^2} \right) = C \rightarrow 2yx^3 + 2y^2x^2 = C$$

16. $y' = (x+y+1)^{1/2} \quad v = x+y+1 \rightarrow y = v-x-1 \quad \frac{dy}{dx} = -1 + \frac{dv}{dx}$

$$-1 + \frac{dv}{dx} = (x+v-x-1+1)^{1/2} \rightarrow \frac{dv}{dx} = \sqrt{v+1} \rightarrow \frac{1}{1+\sqrt{v}} dv = dx \rightarrow \int \frac{1}{1+\sqrt{v}} dv = \int dx$$

$$\int \frac{1}{1+\sqrt{v}} dv \rightarrow \begin{matrix} u = \sqrt{v+1} \\ v = (u-1)^2 \\ dv = 2(u-1) du \end{matrix} \rightarrow \int \frac{2(u-1)}{u} du = \int \frac{2u}{u} - \frac{2}{u} du = 2u - 2 \ln|u| = 2(\sqrt{v+1}) - 2 \ln|\sqrt{v+1}|$$

$$2\sqrt{v+1} - 2 \ln|\sqrt{v+1}| = x + C \rightarrow 2\sqrt{x+y+1} - 2 \ln|1 + \sqrt{x+y+1}| = x + C$$

$$26. y^2 y' + 2xy^3 = 6x \rightarrow y' + 2xy = 6xy^{-2} \quad \begin{cases} P(x) = 2x \\ Q(x) = 6x \end{cases} \quad v = y^{1-2} = y^{-1} \rightarrow y = v^{1/3}$$

$$\frac{dy}{dx} \cdot \frac{1}{3} v^{-2/3} + \frac{1}{3} v^{-2/3} \frac{dv}{dx} \rightarrow \frac{1}{3} v^{-2/3} v' + 2x v^{-1/3} = 6x v^{-2/3} \rightarrow v' + 6xv = 18x \quad \begin{cases} P(x) = 6x \\ Q(x) = 18x \end{cases} \rightarrow p(x) = e^{\int 6x dx} = e^{3x^2}$$

$$e^{3x^2} v' + 6xe^{3x^2} v = 18xe^{3x^2} \rightarrow e^{3x^2} v = \int 18xe^{3x^2} dx \rightarrow e^{3x^2} v = 3e^{3x^2} + C \rightarrow v = 3 + Ce^{-3x^2}$$

$$y^3 = 3 + Ce^{-3x^2}$$

$$8. x^2 y' = xy + x^2 e^{y/x} \rightarrow y' - \frac{1}{x} y = e^{y/x} \rightarrow y = vx \rightarrow v = \frac{y}{x}, y' = v + xv'$$

$$v + xv' = v + e^v \rightarrow xv' = e^v \rightarrow e^{-v} dv = \frac{1}{x} dx \rightarrow \int e^{-v} dv = \int \frac{1}{x} dx \rightarrow -e^{-v} = \ln(x) + C$$

$$e^{-v} = (-\ln(x)) \rightarrow v = -\ln(-\ln(x)) \rightarrow y = -x \ln(-\ln(x))$$

$$14. x^2 y' + 2xy = 5y^3 \rightarrow y' + \frac{2}{x} y = \frac{5}{x^2} y^3 \quad \begin{cases} P(x) = \frac{2}{x} \\ Q(x) = \frac{5}{x^2} \end{cases} \quad v = y^{1-3} = y^{-2} \rightarrow y = v^{-1/2} \quad \frac{dy}{dx} = -\frac{1}{2} v^{-3/2} v'$$

$$-\frac{1}{2} v^{-3/2} v' + \frac{2}{x} v^{-1/2} = \frac{5}{x^2} v^{-3/2} \rightarrow v' - \frac{4}{x} v = -\frac{10}{x^2} \quad \begin{cases} P(x) = -\frac{4}{x} \\ Q(x) = -\frac{10}{x^2} \end{cases} \quad p(x) = e^{\int -4/x dx} = x^{-4}$$

$$x^{-4} v' - \frac{4}{x^5} v = \frac{10}{x^6} \rightarrow x^{-4} v = \int 10x^{-6} dx = 2x^{-5} + C \rightarrow v = \frac{2}{x} + Cx^4 \rightarrow y^2 = \frac{2}{x} + Cx^4 = \frac{(2x^5 + 2)}{x}$$

$$y^2 = \frac{2x}{x^5 + 2}$$

$$42. \frac{2x^{5/2} - 3y^{5/3}}{2x^{5/2} y^{4/3}} dx + \frac{3y^{5/3} - 2x^{5/2}}{3x^{3/2} y^{5/3}} dy = 0 \quad \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}?$$

$$\frac{\partial M}{\partial y} = \frac{-5y^{2/3}(2x^{5/2} y^{2/3}) - 4/3 x^{5/2} y^{-1/3}(2x^{5/2} - 3y^{5/3})}{(2x^{5/2} y^{4/3})^2} = \frac{-10x^{5/2} y^{4/3} - 8/3 x^{5/2} y^{1/3} + 4x^{5/2} y^{4/3}}{4x^5 y^{4/3}} = -\frac{3}{2} x^{-5/2} - \frac{2}{3} y^{-5/3}$$

$$\frac{\partial N}{\partial x} = \frac{(-5x^{3/2})(3x^{3/2} y^{5/3}) - 9/2 x^{1/2} y^{5/3}(3y^{5/3} - 2x^{5/2})}{(3x^{3/2} y^{5/3})^2} = \frac{-15x^3 y^{5/3} - 27/2 x^{1/2} y^{10/3} + 9x^3 y^{5/3}}{9x^3 y^{10/3}} = -\frac{2}{3} y^{-5/3} - \frac{3}{2} x^{-3/2}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \rightarrow \text{Exact}$$

$$\frac{\partial F}{\partial x} = M \rightarrow F = \int \frac{2x^{5/2} - 3y^{5/3}}{2x^{5/2} y^{4/3}} dx = \int \frac{2x^{5/2}}{2x^{5/2} y^{4/3}} - \frac{3y^{5/3}}{2x^{5/2} y^{4/3}} = \int \frac{1}{y^{4/3}} - \frac{3}{2} \frac{y}{x^{5/2}} dx$$

$$= \frac{3}{2} \int \frac{1}{x^{5/2}} = -\frac{3}{2} \int x^{-5/2} dx = -\frac{2}{3} x^{-3/2} - \frac{3y}{2} = -\frac{y}{x^{3/2}} \quad F(x, y) = \frac{2x}{y^{4/3}} + \frac{y}{x^{3/2}} + g(y)$$

$$\frac{\partial F}{\partial y} = 0 \rightarrow -\frac{2}{3} \frac{x}{y^{5/3}} + \frac{1}{x^{3/2}} + g'(y) = \frac{3y^{5/3} - 2x^{5/2}}{3x^{3/2} y^{5/3}} = \frac{1}{x^{3/2}} - \frac{2}{3} \frac{x}{y^{5/3}} \rightarrow g'(y) = 0 \quad g(y) = C$$

$$xy^{-2/3} + yx^{-3/2} = C$$